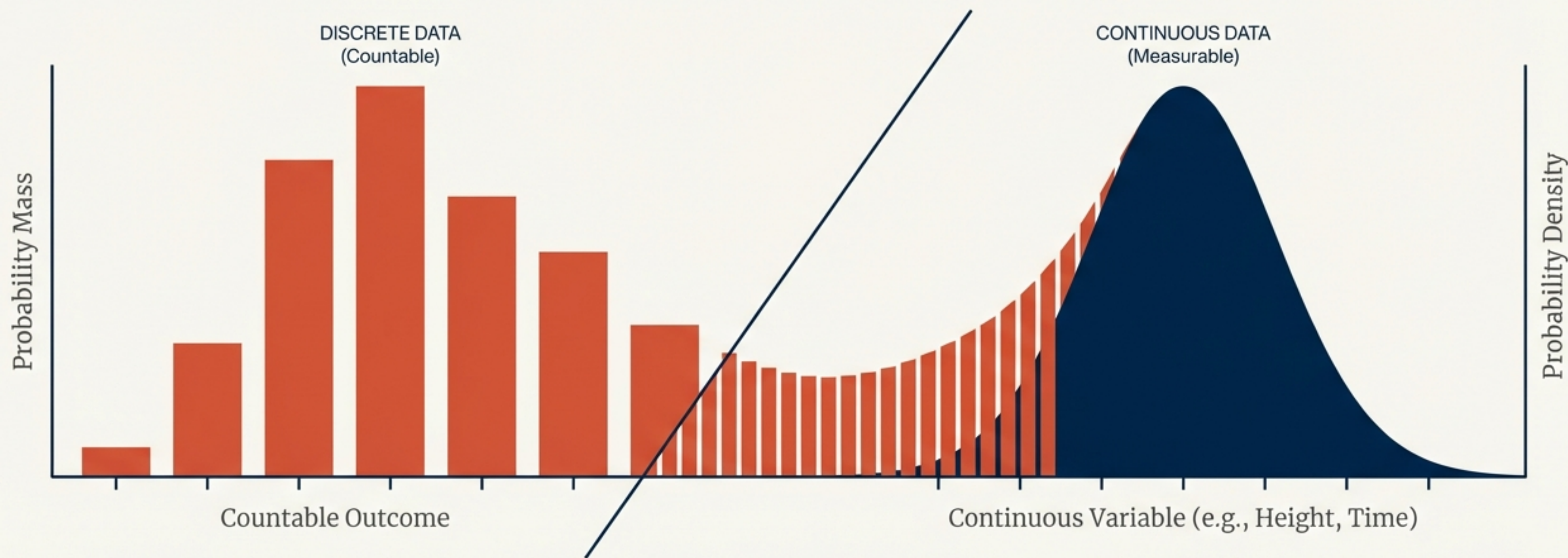


# The Architecture of Randomness: Understanding Statistical Distributions

From Bernoulli Trials to Q-Q Plot Validation

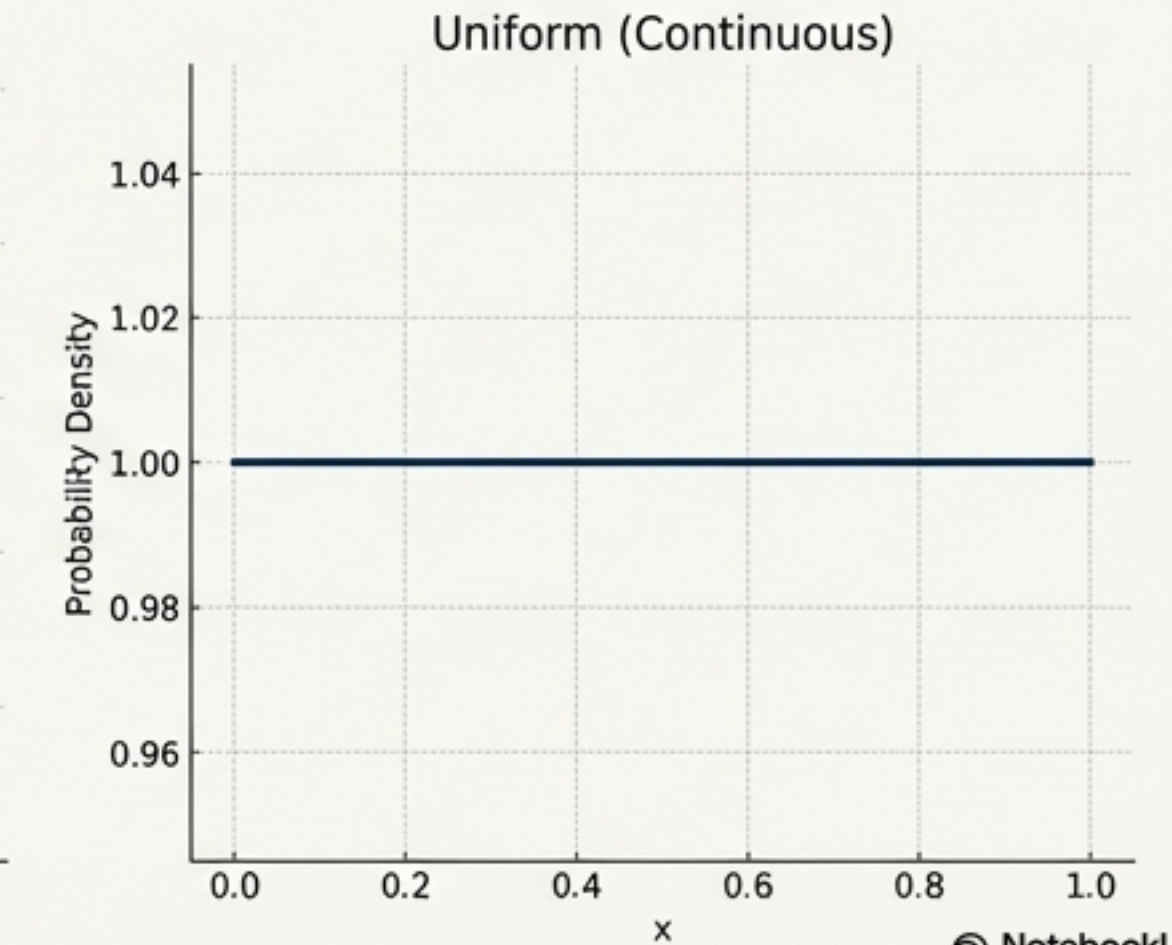
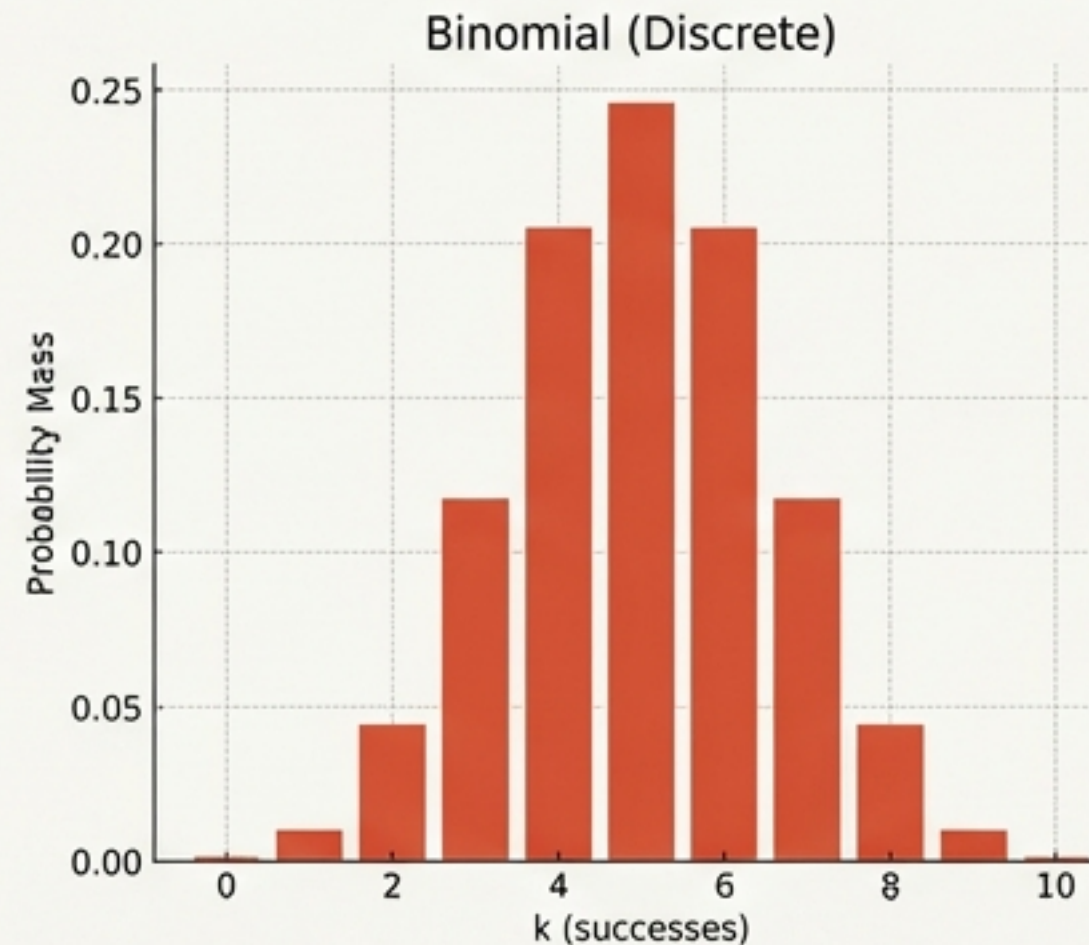
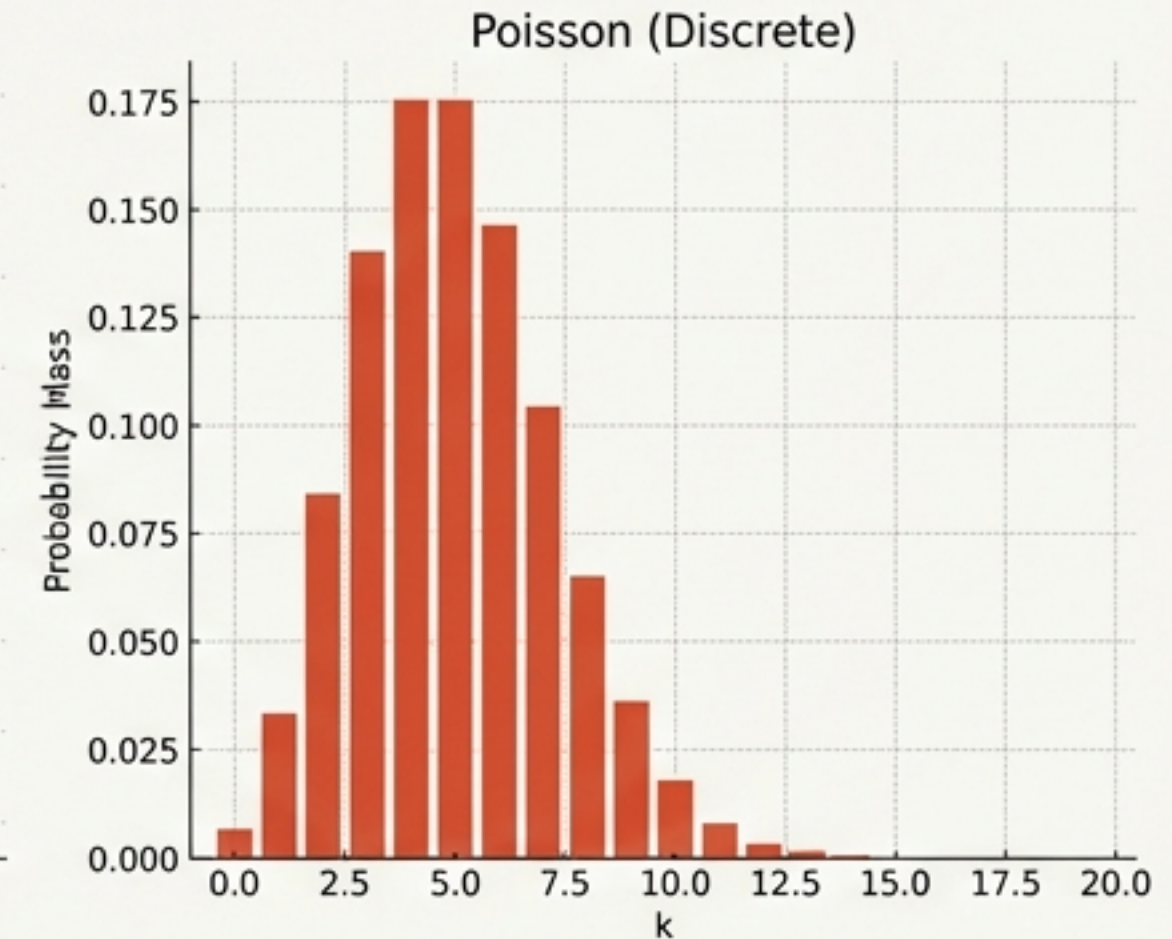
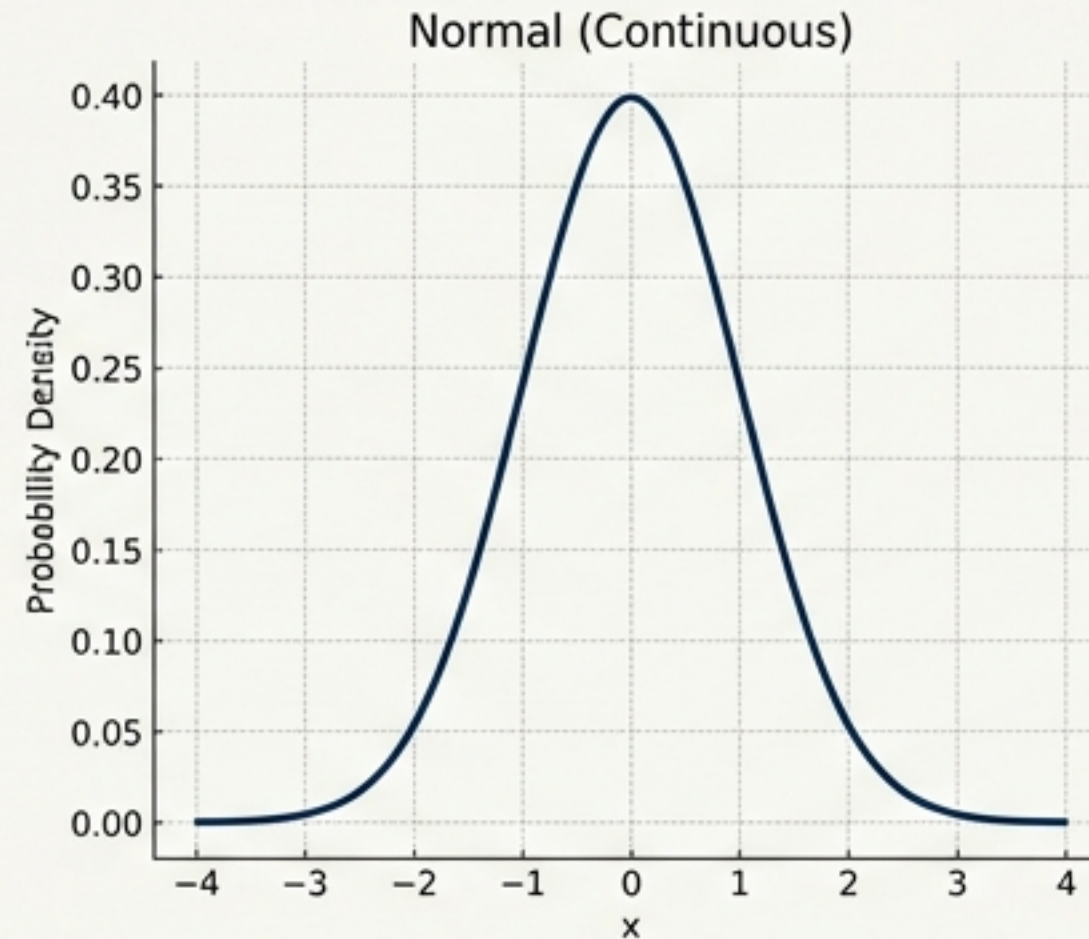


# Distributions Are the Engines of Prediction

A statistical distribution describes how values of a random variable are spread across possible outcomes. It answers two critical questions:

1. What values can the variable take?
2. How often (frequency or probability) do those values occur?

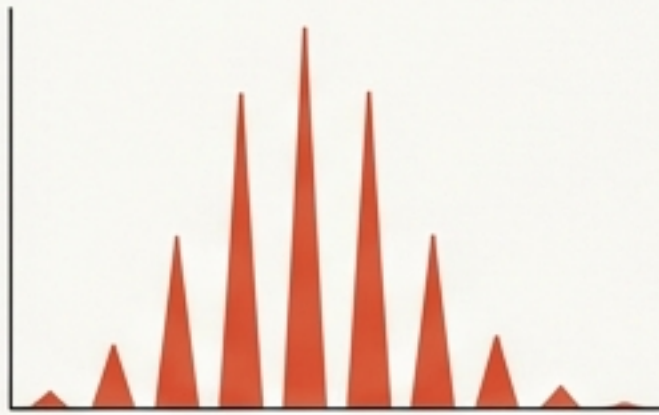
In Data Science, these are not just abstract concepts; they allow us to model real-world phenomena, from stock prices to website traffic, and make probabilistic predictions.



# The Great Divide: Discrete vs. Continuous

## Discrete Distributions

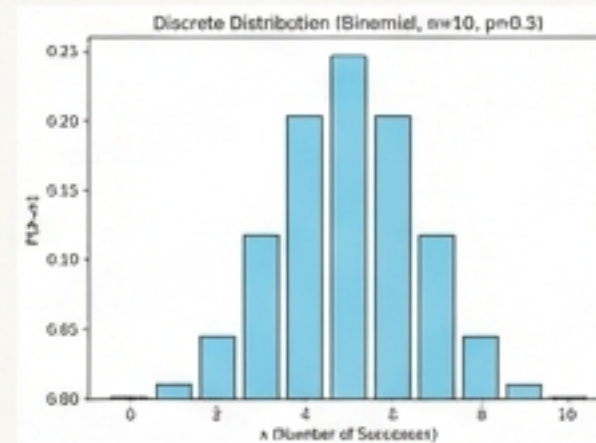
Countable outcomes  
(Finite or step-wise).



Probability Mass Function (PMF)

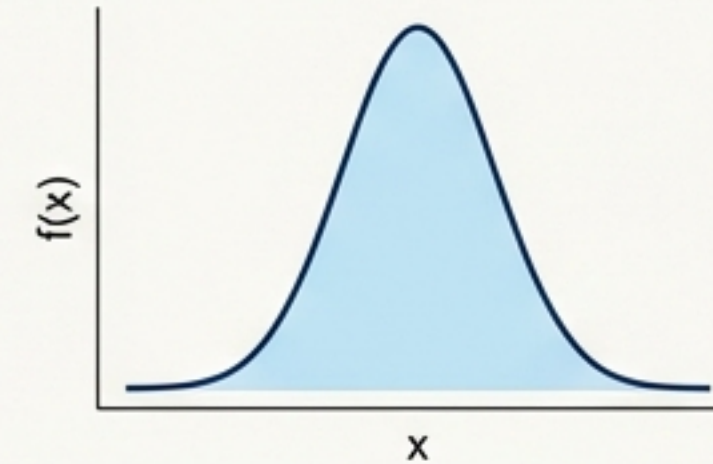
$$P(X = x) \geq 0, \quad \sum P(X = x) = 1$$

Coin tosses (H, T),  
Die rolls (1..6),  
Customer arrivals.



## Continuous Distributions

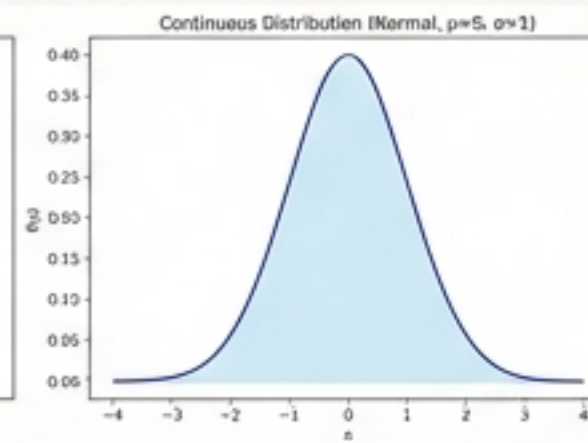
Uncountable outcomes  
(Values within a continuous interval).



Probability Density Function (PDF)

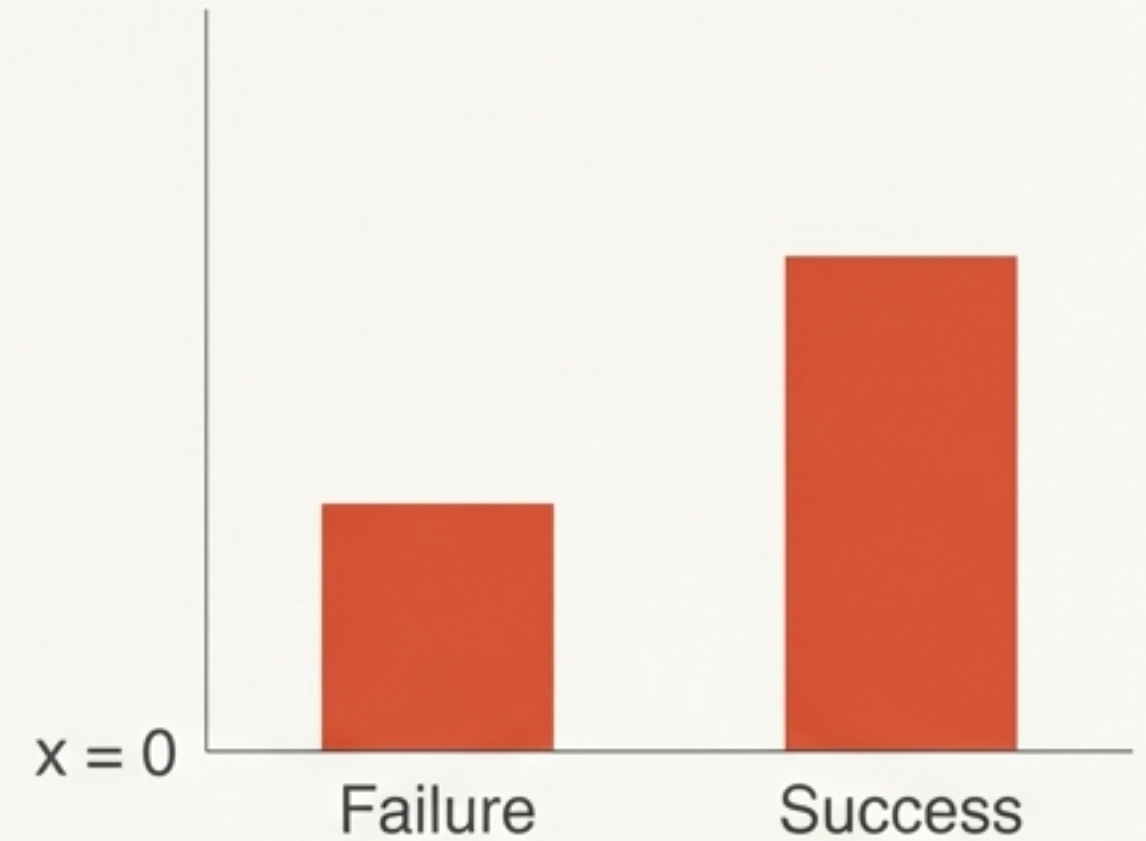
Probability at a specific point is  
0; measured over an interval.

Height (160.2 cm),  
Time taken for a task,  
Stock prices.



# The Atom of Probability: The Bernoulli Distribution

The simplest distribution describing a random experiment with exactly two possible outcomes: Success (1) or Failure (0). Example: A single coin toss or component check.



$$P(X=x) = p^x(1-p)^{1-x}, x \in \{0, 1\}$$

Key Parameters:  $p$  = Probability of success,  
 $1-p$  = Probability of failure

Mean (Expected Value):  $E[X] = p$

Variance:  $\text{Var}(X) = p(1-p)$

# Scaling Up: The Binomial Distribution

The Binomial distribution models the number of successes ( $k$ ) in  $n$  independent Bernoulli trials.

Example: Predicting how many customers out of 100 will buy a product if probability of purchase is 0.3.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- $n$  = number of trials
- $k$  = number of successes
- $p$  = probability of success in one trial
- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Mean:  $E[X] = np$

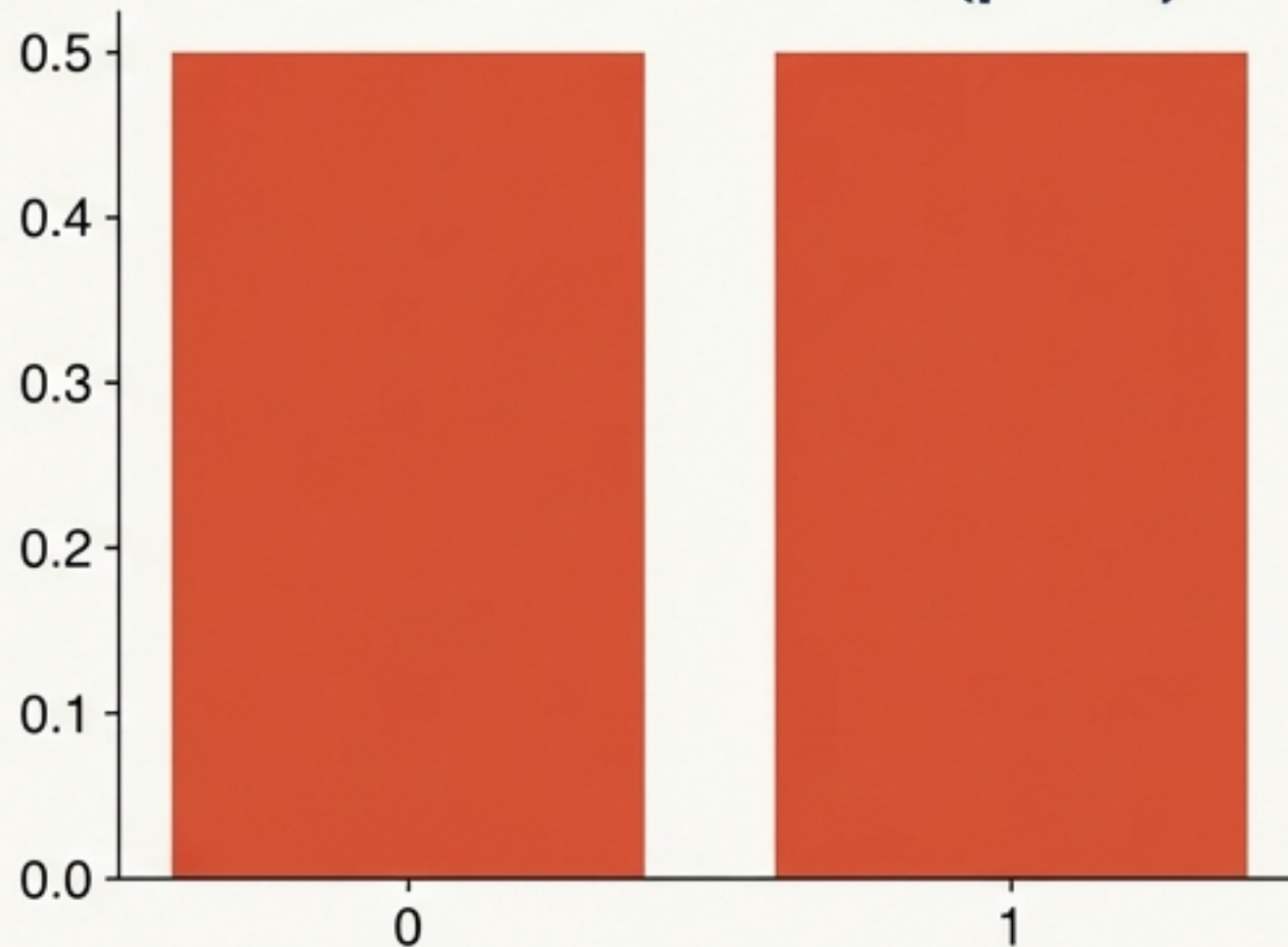
Variance:  $\text{Var}(X) = np(1 - p)$

# Visualising the Relationship

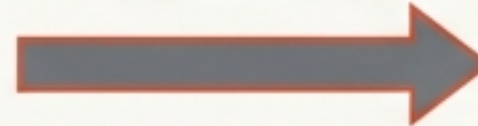
The Bernoulli distribution is a Binomial distribution where  $n=1$ .

$$\text{Bernoulli}(p) = \text{Binomial}(n=1, p)$$

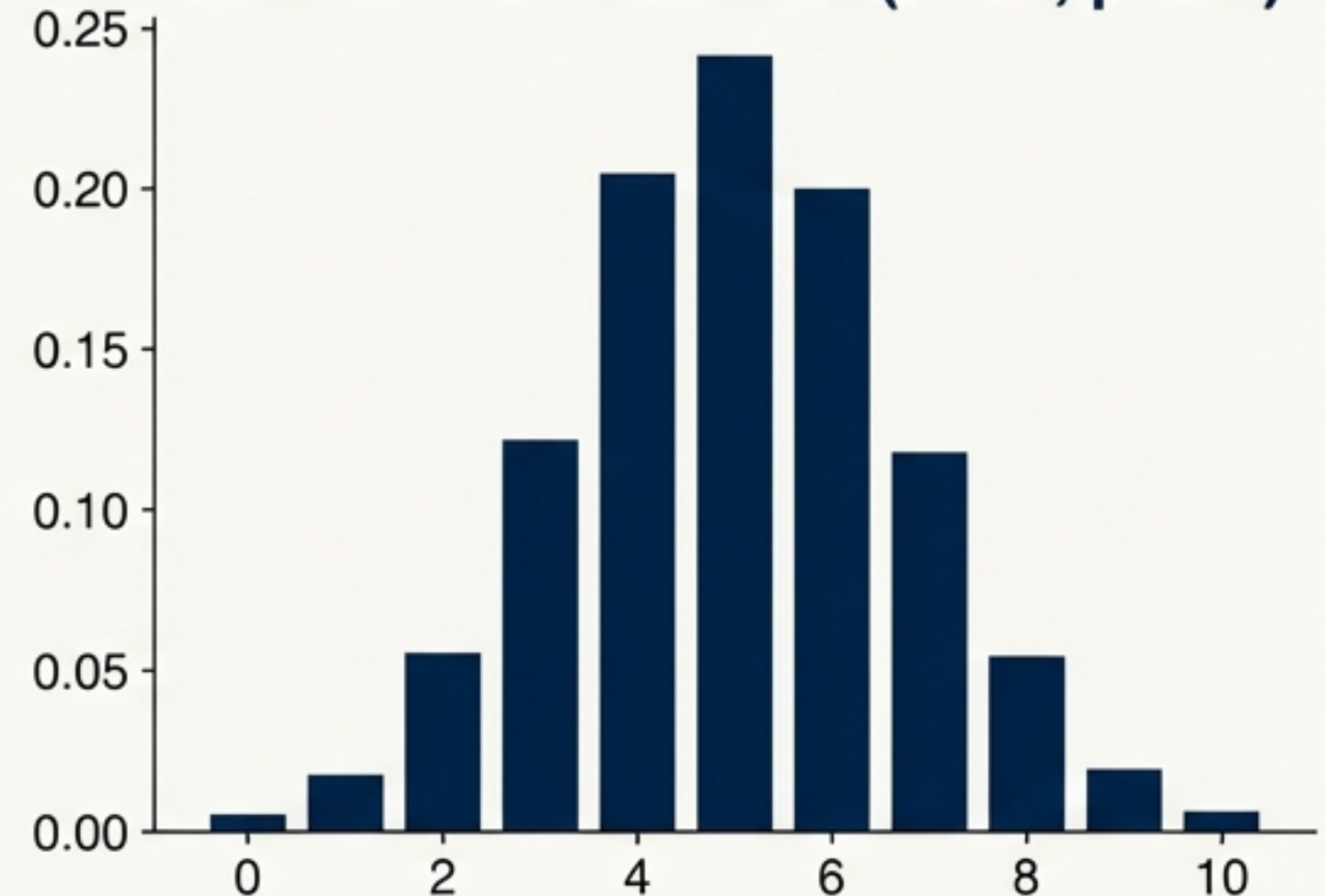
**Bernoulli Distribution ( $p=0.5$ )**



Repeating the  
single trial  $n$  times  
creates a distribution  
of outcomes.



**Binomial Distribution ( $n=10, p=0.5$ )**

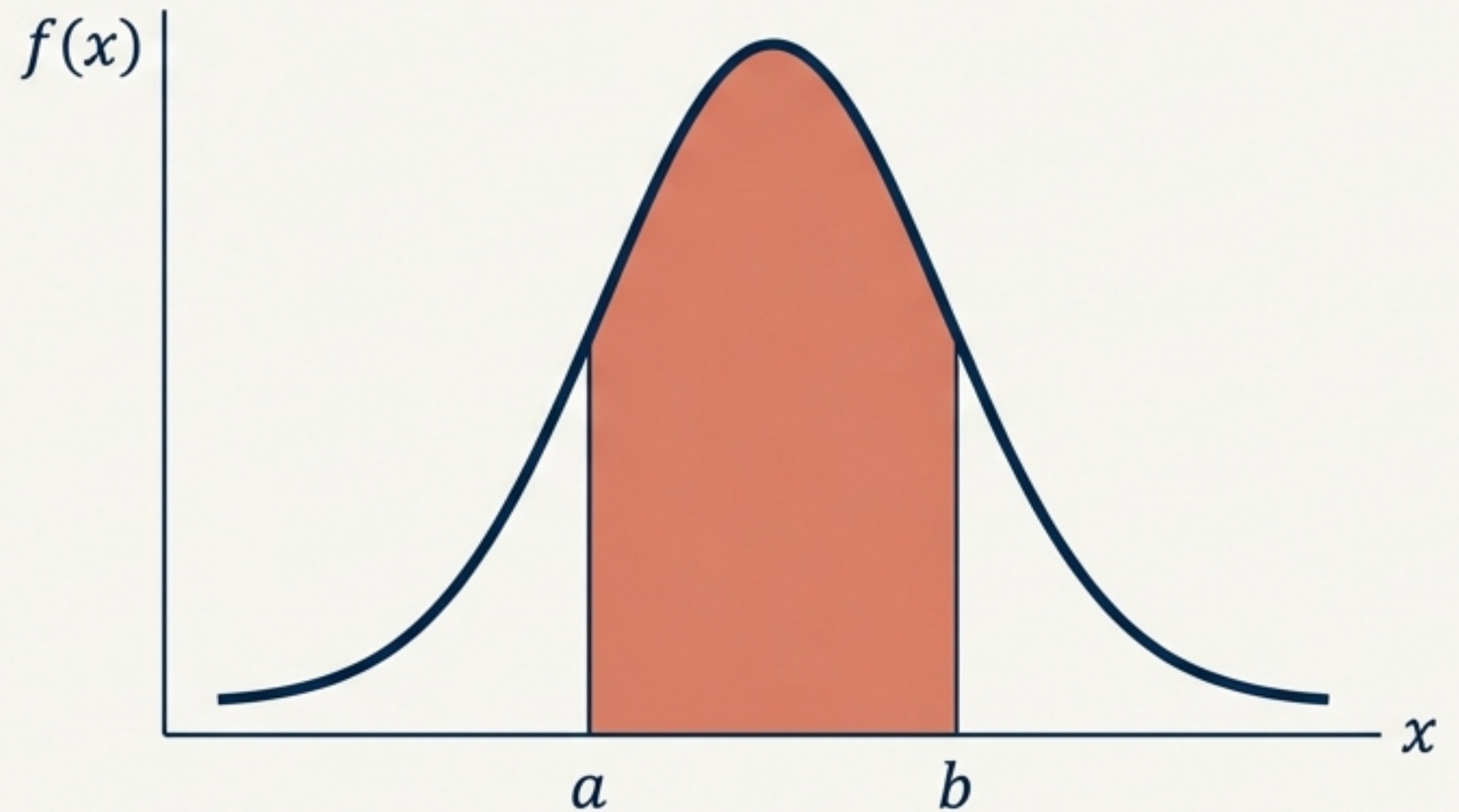


# Continuous Distributions & The Area Under the Curve

When variables take infinite values (like time or distance), we cannot assign probability to a single point. Instead, we measure probability over an interval.

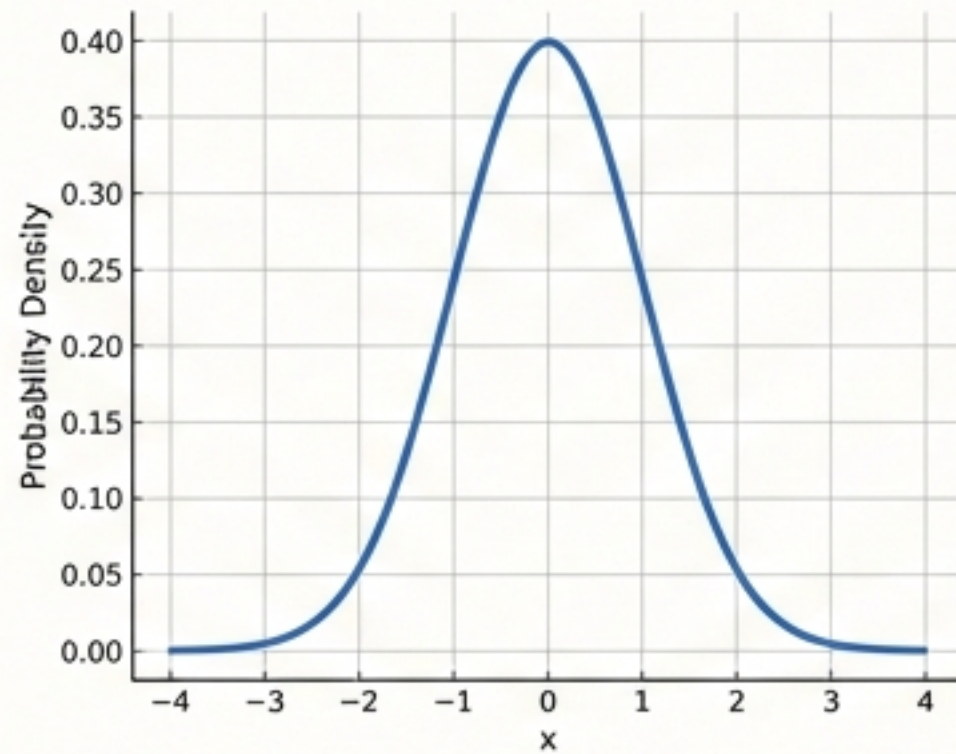
The total area under the curve always equals 1.

$$P(a \leq X \leq b) = \int_a^b f(x) dx, \quad \int_{-\infty}^{\infty} f(x) dx = 1$$



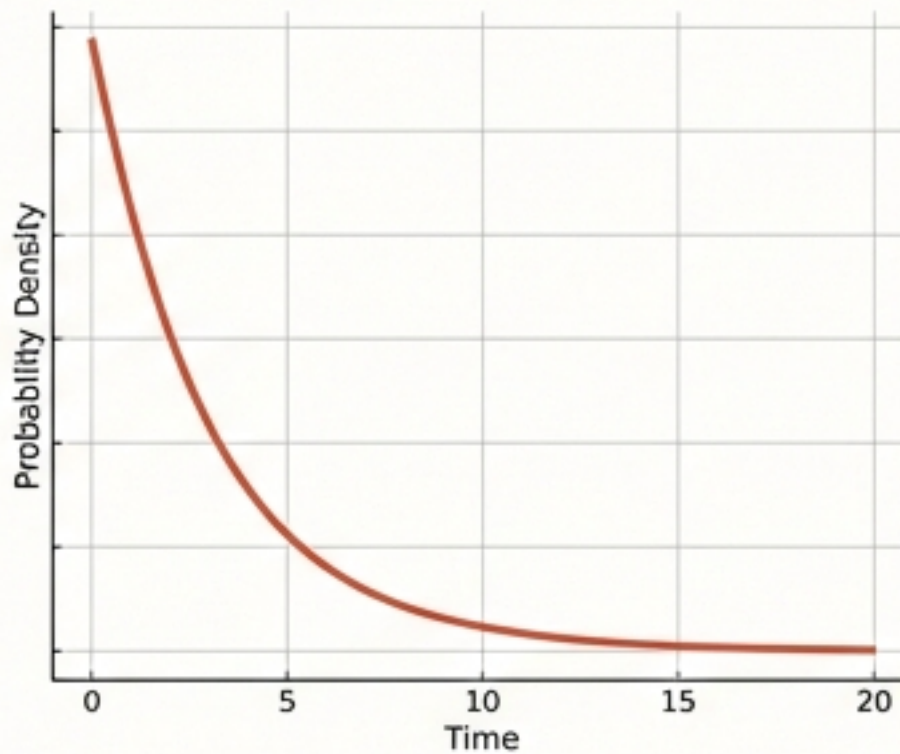
# The Continuous Gallery: Common Shapes

## Normal (Gaussian)



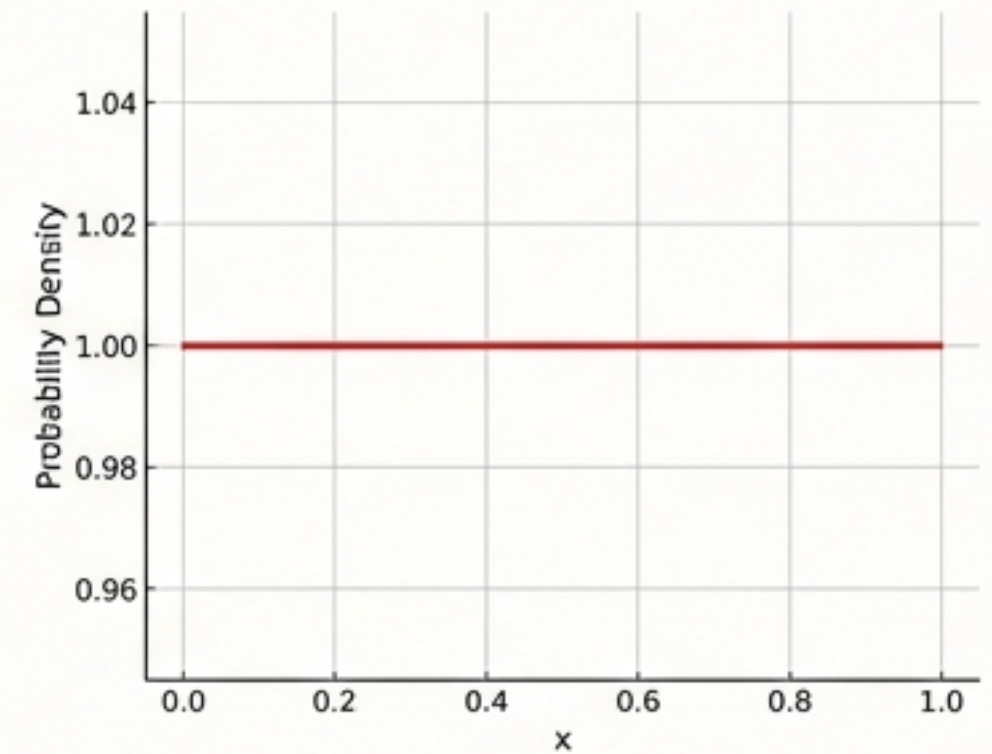
Most naturally occurring data (heights, test scores, measurement errors).

## Exponential



Time between events (e.g., time between bus arrivals or server requests).

## Uniform



Equal likelihood across a range (e.g., random number generator).

# Implementation in Python

Using `scipy.stats` for Discrete and Continuous models

## Discrete: Using PMF

```
from scipy.stats import binom
n, p = 10, 0.5
x = np.arange(0, n+1)
# PMF is used for Discrete
y = binom.pmf(x, n, p)
plt.bar(x, y)
```

Note the method switch:  
.pmf() generates bars for discrete data, while .pdf() generates curves for continuous data.

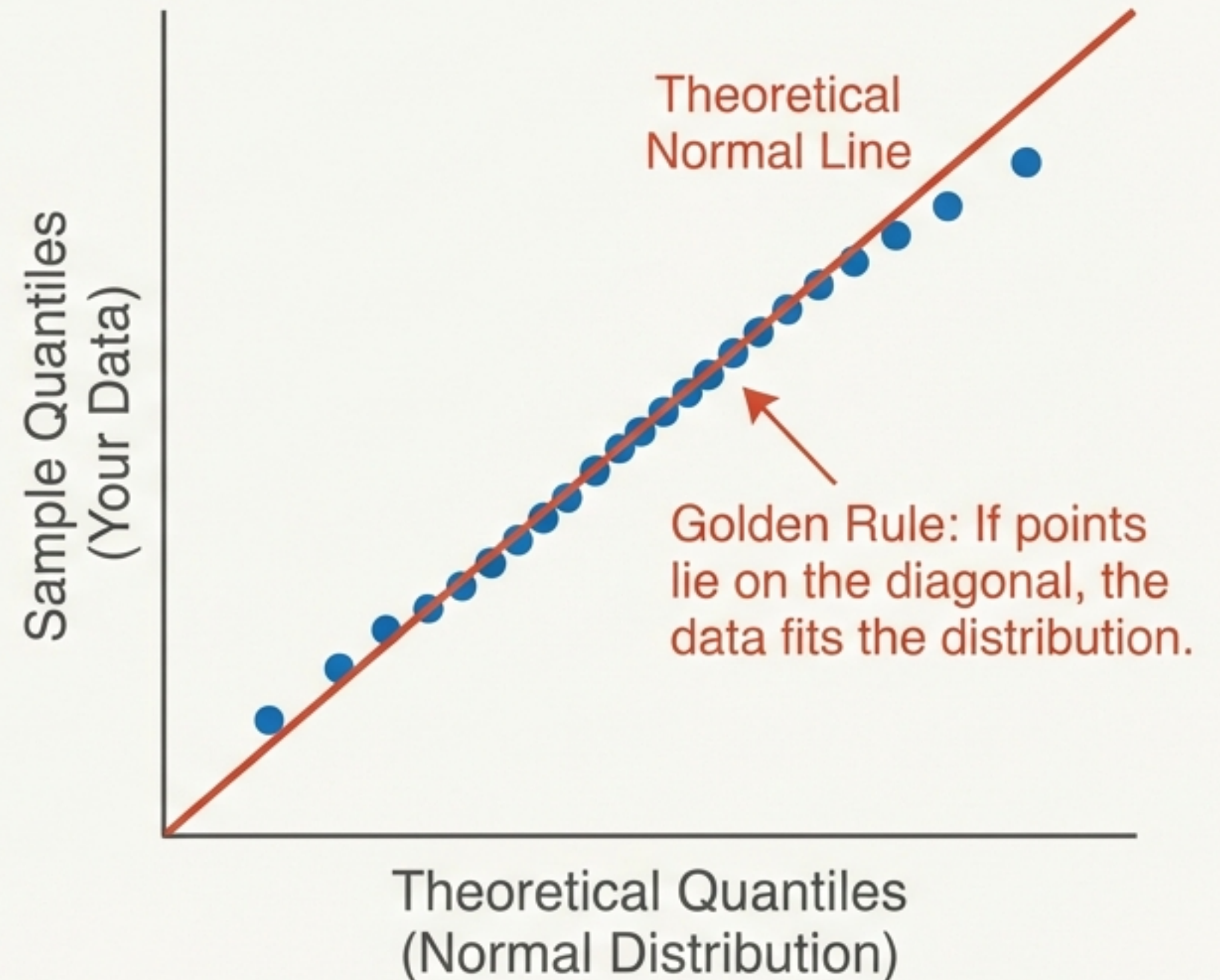
## Continuous: Using PDF

```
from scipy.stats import norm
x = np.linspace(-4, 4, 1000)
# PDF is used for Continuous
y = norm.pdf(x, 0, 1)
plt.plot(x, y)
```

# Validating Assumptions: The Q-Q Plot

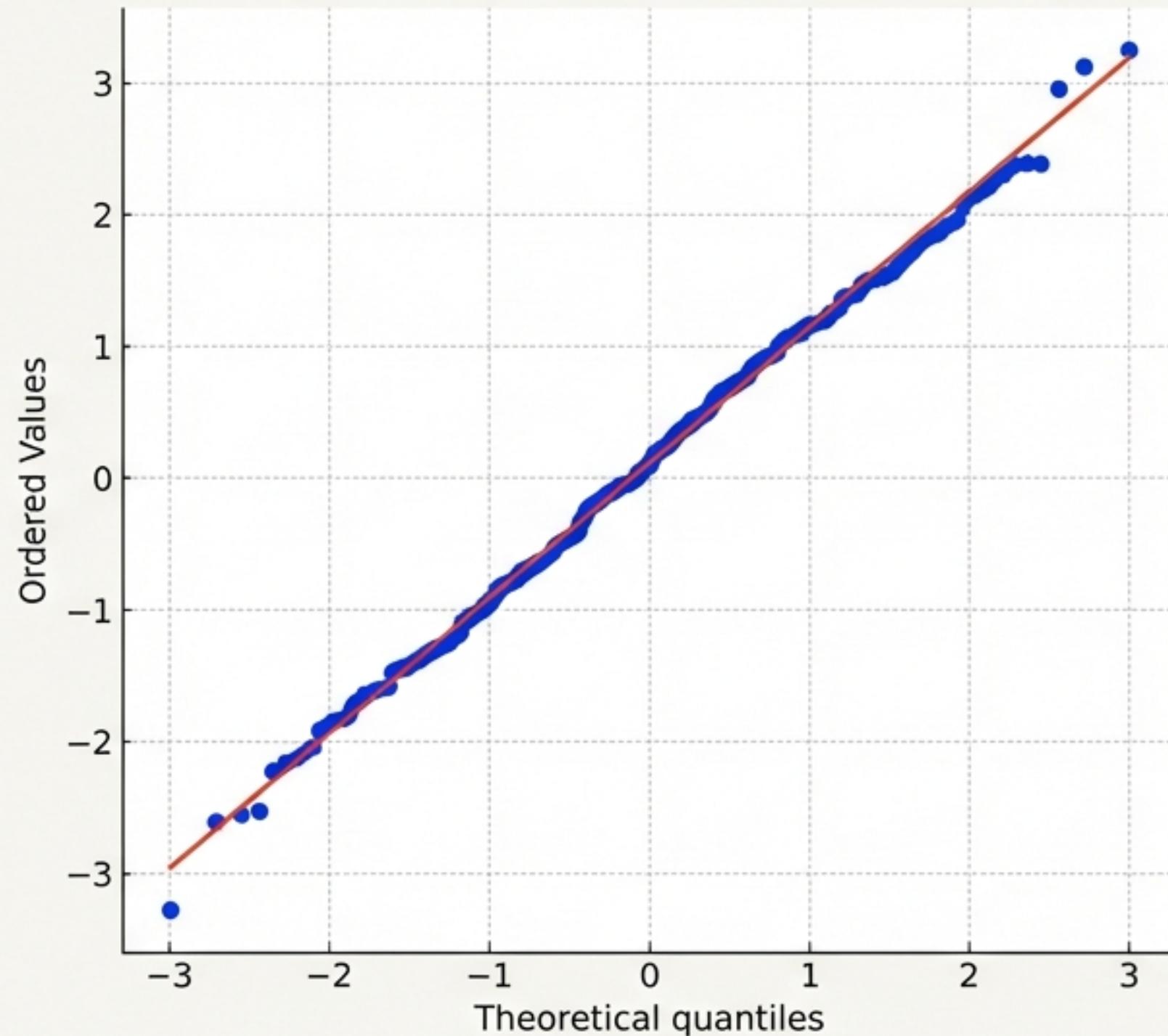
A Q-Q (Quantile-Quantile) plot is a graphical tool to check if a dataset follows a theoretical distribution (usually Normal).

Why it matters: Many statistical models assume normality; this is how we verify that assumption.



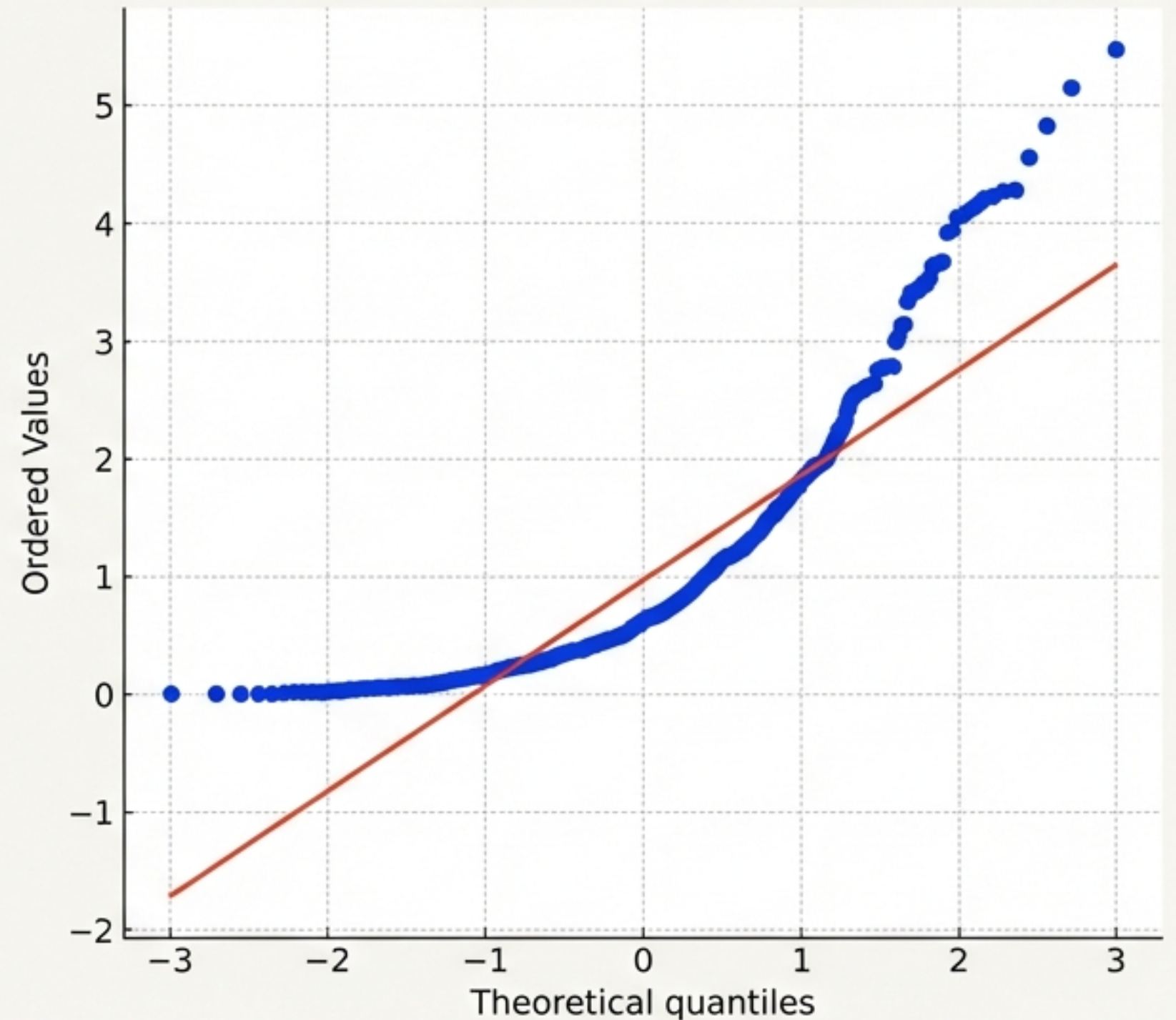
# Reading the Q-Q Plot: A Diagnostic Test

## Scenario A: Good Fit (Normal Data)



Points follow the diagonal line.  
Conclusion: Data is approximately Normally distributed.

## Scenario B: The Mismatch (Exponential vs Normal)



Points curve away from the line.  
Conclusion: Data is NOT Normal.

# The Statistical Toolkit

## 1

### Categorise

Determine if your data is Discrete (Bernoulli, Binomial, Poisson) or Continuous (Normal, Exponential, Uniform).

## 2

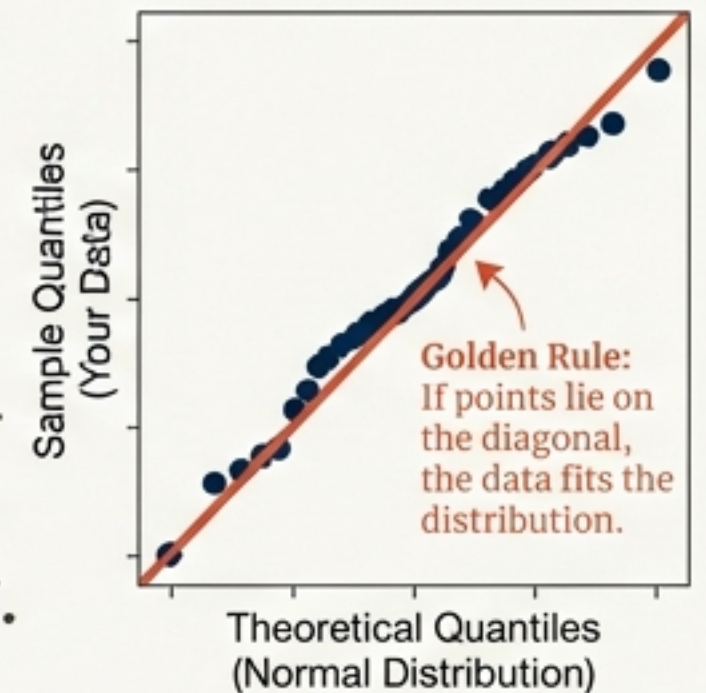
### Model

Use PMF for discrete probabilities (sum to 1). Use PDF and integration for continuous probabilities (area under curve).

## 3

### Validate

Deploy Q-Q Plots to test your distribution assumptions. Look for the diagonal alignment to confirm normality.



Understanding the distribution is the first step to modelling the reality.